Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | ***Time: 03 Hours***
Code: 101617 | ***Subject: Concrete Materials*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) C_3S in cement mainly contributes to
(i) Long-term strength (iii) Initial setting and early strength
(ii) Soundness (iv) Colouring

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(b) Hydration of cement produces calcium silicate hydrate and
(i) Sodium oxide (iii) Calcium hydroxide
(ii) Ferric oxide (iv) Magnesium chloride

(c) M-sand provides better performance than natural sand because of
(i) Rounded shape (iii) Uniform grading and angularity
(ii) High porosity (iv) High silt content

(d) Segregation during placement is reduced by
(i) Dropping concrete from greater height (iii) Using cohesive mixes
(ii) High water content (iv) Over-vibration

(e) Sulphate attack on concrete mainly deteriorates
(i) C–S–H (iii) Aggregates
(ii) Calcium aluminates (iv) Reinforcement

(f) Modulus of elasticity of concrete depends primarily on
(i) Curing temperature (iii) Elastic properties of aggregates
(ii) Compaction method (iv) Cement colour

(g) Aggregates with higher flakiness index tend to
(i) Improve workability (iii) Reduce strength
(ii) Increase durability (iv) Reduce water demand

(h) M-sand sample has bulk density = 1600 kg/m³. Volume required for 800 kg is
(i) 0.40 m³ 0.50 m³
(ii) 0.60 m³ 0.48 m³

(i) Hooked-end steel fibres offer improvement mainly in
(i) Air content (iii) Bond and pull-out resistance
(ii) Bleeding (iv) Flowability

(j) A cement sample gives the following sieving result: Retained on 90-micron sieve = 8%. The fineness of cement is
(i) Within limit (iii) Marginal
(ii) Outside limit (iv) Not identifiable

Q.2 (a) Explain the chemical composition of Ordinary Portland Cement (OPC) and describe the role of each compound (C_3S, C_2S, C_3A, C_4AF) in hydration. **[7]**

(b) Discuss the factors that influence the rate of hydration in cement and their effects on early-age strength development. **[7]**

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Q.3 (a) Describe the physical properties of aggregates as per IS 2386 and their significance in performance of concrete. [7]

(b) Explain the concept of grading curve and discuss how poorly graded aggregates affect workability and strength. [7]

Q.4 (a) What are the advantages of Fibre Reinforced Concrete (FRC) and describe the behaviour of concrete using steel fibres. [7]

(b) A mix contains fibres at 0.4% by volume. Calculate the fibre volume required for 1.5 m^3 of concrete. [7]

Q.5 (a) Define Self-Compacting Concrete (SCC) and explain the essential characteristics: filling ability, passing ability, and resistance to segregation. [7]

(b) Describe the typical materials used in SCC mix design and the role of viscosity-modifying admixtures. [7]

Q.6 (a) Write a detailed note on High Performance Concrete (HPC) and highlight its distinguishing features compared to conventional concrete. [7]

(b) Explain the role of supplementary cementitious materials in achieving low permeability and high durability in HPC. [7]

Q.7 (a) Discuss the steps involved in concrete mix design according to IS 10262: 2019, including considerations for special concretes. [7]

(b) Discuss the behaviour of concrete under fatigue, impact, and sustained loading, with examples of structural applications. [7]

Q.8 (a) Describe the process of placing, compacting, and finishing concrete to ensure minimum segregation and maximum density. [7]

(b) Explain the effect of inadequate vibration on strength and durability of hardened concrete. [7]

Q.9 Write short note on ***any four*** of the following: **[3.5 x 4 = 14]**

(a) High volume fly ash concrete
(b) Methods of controlling sulphate attack
(c) Pullout test
(d) Polymer concrete
(e) Fill box test
(f) Orimet test

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